

16 Functions - Zero to Hero questions list

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1 Functions in C — Practice Questions (50 Questions with Hints Only)

The questions are arranged from **Beginner** → **Intermediate** → **Advanced**.

2 Beginner Level (1–15)

2.0.1 1. Write a function to print “Hello, World!”

Hint: Function with no parameters and no return value.

2.0.2 2. Write a function to print your name 10 times.

Hint: Use a loop inside the function.

2.0.3 3. Write a function that takes an integer and prints it.

Hint: Pass one parameter.

2.0.4 4. Write a function that returns the square of a number.

Hint: Return $n * n$.

2.0.5 5. Write a function to add two numbers.

Hint: Pass two integers and return their sum.

2.0.6 6. Write a function to subtract two numbers.

Hint: Return $a - b$.

2.0.7 7. Write a function to multiply two numbers.

Hint: Return $a * b$.

2.0.8 8. Write a function to divide two numbers.

Hint: Check for division by zero.

2.0.9 9. Write a function to find the larger of two numbers.

Hint: Use an if statement.

2.0.10 10. Write a function to find the smaller of two numbers.

Hint: Compare two integers.

2.0.11 11. Write a function that returns whether a number is even or odd.

Hint: Use $\% 2$.

2.0.12 12. Write a function that returns the cube of a number.

Hint: Multiply the number three times.

2.0.13 13. Write a function to calculate the area of a rectangle.

Hint: $\text{length} \times \text{breadth}$.

2.0.14 14. Write a function to calculate the perimeter of a rectangle.

Hint: $2 \times (\text{length} + \text{breadth})$.

2.0.15 15. Write a function to swap two numbers using a third variable.

Hint: Pass pointers.

3 Intermediate Level (16–35)

3.0.1 16. Write a function to find the maximum of three numbers.

Hint: Compare step by step.

3.0.2 17. Write a function to find the minimum of three numbers.

Hint: Use nested comparisons.

3.0.3 18. Write a function to calculate the factorial of a number.

Hint: Use a loop.

3.0.4 19. Write a recursive function to calculate factorial.

Hint: Base case: $0! = 1$.

3.0.5 20. Write a function to calculate the power of a number.

Hint: Multiply repeatedly.

3.0.6 21. Write a recursive function for power.

Hint: Reduce exponent each call.

3.0.7 22. Write a function to check whether a number is prime.

Hint: Test divisibility from 2 to $\sqrt{n}$.

3.0.8 23. Write a function to print all prime numbers between two numbers.

Hint: Call the prime-check function repeatedly.

3.0.9 24. Write a function to reverse a number.

Hint: Extract digits using % 10.

3.0.10 25. Write a function to check whether a number is a palindrome.

Hint: Compare original and reversed number.

3.0.11 26. Write a function to count digits in a number.

Hint: Divide by 10 repeatedly.

3.0.12 27. Write a function to find the sum of digits.

Hint: Extract each digit.

3.0.13 28. Write a function to check whether a number is an Armstrong number.

Hint: Sum powers of digits.

3.0.14 29. Write a function to calculate GCD of two numbers.

Hint: Use Euclid's Algorithm.

3.0.15 30. Write a function to calculate LCM of two numbers.

Hint: Use GCD.

3.0.16 31. Write a function to generate Fibonacci series.

Hint: Use iteration.

3.0.17 32. Write a recursive Fibonacci function.

Hint: $F(n) = F(n-1) + F(n-2)$.

3.0.18 33. Write a function to find the sum of first N natural numbers.

Hint: Use a loop or formula.

3.0.19 34. Write a function to find the average of an array.

Hint: Pass array and size.

3.0.20 35. Write a function to find the largest element in an array.

Hint: Traverse once.

4 Arrays and Strings with Functions (36–45)

4.0.1 36. Write a function to find the smallest element in an array.

Hint: Keep updating minimum.

4.0.2 37. Write a function to count even numbers in an array.

Hint: Check % 2.

4.0.3 38. Write a function to count odd numbers in an array.

Hint: Opposite of even.

4.0.4 39. Write a function to calculate the sum of array elements.

Hint: Traverse once.

4.0.5 40. Write a function to reverse an array.

Hint: Swap first and last elements.

4.0.6 41. Write a function to search an element in an array.

Hint: Use Linear Search.

4.0.7 42. Write a function to sort an array in ascending order.

Hint: Use Bubble Sort.

4.0.8 43. Write a function to calculate the length of a string.

Hint: Traverse until '\0'.

4.0.9 44. Write a function to count vowels in a string.

Hint: Compare each character.

4.0.10 45. Write a function to reverse a string.

Hint: Swap characters from both ends.

5 Advanced Level (46–50)

5.0.1 46. Write a function that swaps two numbers without using a third variable.

Hint: Use arithmetic or XOR.

5.0.2 47. Write a function that accepts another function using a function pointer.

Hint: Learn function pointer syntax.

5.0.3 48. Write a recursive function to solve the Tower of Hanoi problem.

Hint: Move $n-1$ disks recursively.

5.0.4 49. Write a menu-driven calculator using functions.

Hint: Create separate functions for each operation and call them from `switch`.

5.0.5 50. Build a mini Student Record Management System using functions.

Hint: Divide the program into functions like:

- Add Student

- Display Students
- Search Student
- Update Student
- Delete Student
- Exit